

Classwork Problems

Unit 2: Differentiation and its Applications

L17, L18: Second order derivatives

Let $y = f(x)$ be a function.

Then $y_1 = \frac{dy}{dx} = f'(x)$ is called the first order derivative.

$y_2 = \frac{d^2y}{dx^2} = f''(x)$ is called the second order derivative.

$y_3 = \frac{d^3y}{dx^3} = f'''(x)$ is called the third order derivative.

In general n^{th} order derivative of $y = f(x)$ is denoted by $y_n = \frac{d^ny}{dx^n} = f^n(x)$

Find $\frac{d^2y}{dx^2}$ for the following

(1) $y = x^2 + 3x + 4$

Ans.2

$$\frac{dy}{dx} = 2x + 3$$

Differentiate w.r.t. x

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (2x + 3)$$

$$\frac{d^2y}{dx^2} = 2$$

(2) $y = x^5 + \sin x$

Ans. $20x^3 - \sin x$

(3) $y = \log x$

Ans. $-\frac{1}{x^2}$

(4) $y = e^x \sin 5x$

Ans. $2e^x (5 \cos 5x - 12 \sin 5x)$

(5) $y = x \cdot \cos x$

Ans. $-x \cos x - 2 \sin x$

(6) $y = \log(\log x)$

Ans. $\frac{-(1 + \log x)}{(x \log x)^2}$

(7) If $y = e^{2x+3}$, show that $\frac{d^2y}{dx^2} = 4y$

(8) If $y = \sin^{-1} x$, show that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 0$

(9)

If $y = A \sin x + B \cos x$, then prove that $\frac{d^2y}{dx^2} + y = 0$.

(10)

If $e^y(x+1) = 1$, show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$

L19-L21: Applications of derivatives

(i) Rate of change

- (1) Find the rate of change of the area of a circle with respect to its radius r when $r = 3$ cm.
- (2) An edge of a variable cube is increasing at the rate of 3cm/s. How fast is the volume of the cube increasing when the edge is 10cm long?
- (3) For the curve $y = 5x - 2x^3$, if x increases at the rate of 2 units/sec, then how fast is the slope of the curve changing when $x = 3$?
- (4) For what values of x is the rate of increase of $x^3 - 5x^2 + 5x + 8$ is twice the rate of Increase of x ?
- (5) The volume of a sphere is increasing at the rate of 20 cubic centimetres per second. Find the rate of change of its surface area at the instant when its radius is 8 cm.
- (6) Water is dripping out from a conical funnel of semi-vertical angle $\frac{\pi}{4}$ at the uniform rate of $2 \text{ cm}^2/\text{sec}$ in the surface area, through a tiny hole at the vertex of the bottom. When the slant height of cone is 4 cm, find the rate of decrease of the slant height of water.

(ii) Increasing/decreasing functions

- (1) Show that the function $f(x) = x^3 - 6x^2 + 15x + 3$ is increasing for all $x > 0$.
- (2) For what values of x is $f(x) = 2x^3 - 9x^2 + 12x + 4$, is a decreasing function?
Ans: $f(x)$ is decreasing for all $x \in (1, 2)$.
- (3) Show that the function $f(x) = x^3 + 9x^2 + 30x + 7$ is always increasing.
- (4) State the intervals on which $f(x)$ is increasing or decreasing, where
 $f(x) = x^3 - 12x + 1$.
Ans: $f(x)$ is increasing for all $x \in (-\infty, -2), x \in (2, \infty)$ and decreasing for all $x \in (-2, 2)$.

(iii) Maxima & Minima

- (1) Determine the maximum and minimum values of the function
 $f(x) = 2x^3 - 21x^2 + 36x - 20$
Ans: $f(x)$ has minima at $x = 6$ and $f_{\min} = -128$. $f(x)$ has maxima at

$$x = 1, f_{\max} = -3$$

- (2) Determine the maximum and minimum values of the function

$$f(x) = 5x^6 + 18x^5 + 15x^4 - 10.$$

Ans: $f(x)$ has minima at $x = -2$ and $f_{\min} = -26$.

$f(x)$ has maxima at $x = -1, f_{\max} = -8$.

- (3) Determine the maximum and minimum values of the function

$$f(x) = x^3 - 9x^2 + 24x - 5$$

Ans: $f(x)$ has minima at $x = 4$ and $f_{\min} = 11$.

$f(x)$ has maxima at $x = 2, f_{\max} = 15$

- (4) Determine the maximum and minimum values of the function

$$f(x) = 10x^6 - 24x^5 + 15x^4 - 40x^3 + 108$$

Ans: $f(x)$ has minima at $x = 2$ and $f_{\max} = -100$.

Additional Problems

(1) Find $\frac{d^2y}{dx^2}$, if $y = x^3 - 5x + 1$

Ans. $6x$

(2) Find $\frac{d^2y}{dx^2}$, if $y = x^3 + \tan x$

Ans. $6x + 2\sec^2 x \cdot \tan x$

(3) If $y = 3e^{2x} + 2e^{3x}$, prove that $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$

(4) If $y = 2\sin 3x - 5\cos 3x$, show that $\frac{d^2y}{dx^2} + 9y = 0$

(5) Find $\frac{d^2y}{dx^2}$, if $y = e^{10x} + 10x$

Ans. $100e^{10x}$

(6) The side of a square is increasing at the rate of 0.02 cm/sec. What is the rate at which the area is increasing, when the side of the square is 6 centimetres?

(7) A kite is moving horizontally at a height of 151.5 meters. If the speed of kite is 10 m/s, how fast is the string being let out; when the kite is 250 m away from the boy who is flying the kite? The height of boy is 1.5 m.

(8) Water is dripping out at a steady rate of 1 cu cm/sec through a tiny hole at the vertex of the conical vessel, whose axis is vertical. When the slant height of water in the vessel is 4 cm, find the rate of decrease of slant height, where the vertical angle of the conical vessel is $\frac{\pi}{6}$.

- (9) If the area of a circle increases at a uniform rate, then prove that perimeter varies inversely as the radius.
- (10) A spherical ball of salt is dissolving in water in such a manner that the rate of decrease of the volume at any instant is proportional to the surface. Prove that the radius is decreasing at a constant rate.
- (11) Water is dripping out at a steady rate of 1 cu cm/sec through a tiny hole at the vertex of the conical vessel, whose axis is vertical. When the slant height of water in the vessel is 4 cm, find the rate of decrease of slant height, where the vertical angle of the conical vessel is $\frac{\pi}{6}$.
- (12) For $f(x) = x^4 - 8x^2$, determine all intervals where f is increasing or decreasing.

Ans. f is increasing in $(-2, 0)$ and $(2, \infty)$ & f is decreasing in $(-\infty, -2)$ and $(0, 2)$

- (13) Determine the maximum and minimum values of the function

$$f(x) = x^3 - 3x + 4$$

Ans. Max value = 22 at $x = 3$ and min value = -14 at $x = -3$